👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 7 · STANDARD 6.AT.8

Equations and Inequalities Problem Solving

Lesson 7-7 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes **Level 1 Support**

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can model and solve real-world problems using equations and inequalities.

Language Objective: I can explain my reasoning using the words model, equation, inequality, and reasonableness.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Model **Equation** **Inequality** **Reasonableness** **Variable** **Constraint**

👆 *Tap any word to see what it means and a picture.*

1. A drawing, equation, or diagram that represents a situation is a ____________.

2. A math sentence with an equal sign is an ____________.

3. A math sentence that compares values with <, >, ≤, or ≥ is an ____________.

4. Checking whether an answer makes sense for the problem is checking its ____________.

5. A letter that stands for the unknown amount in a problem is a ____________.

6 A limit that tells which values are allowed in a problem is a ____________________ .

Reflect — Exit Ticket

A box of donuts has some donuts. After giving away 7, there are fewer than 5 left. Which inequality represents the starting number of donuts d?

A. $d - 7 < 5$

B. $d + 7 < 5$

C. $d - 7 > 5$

D. $d - 7 = 5$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Model
2. **Notes 2:** Equation
3. **Notes 3:** Inequality
4. **Notes 4:** Reasonableness
5. **Notes 5:** Variable
6. **Notes 6:** Constraint
7. **Watch for this mistake:** *A common mistake in Equations and Inequalities Problem Solving is skipping the key idea: "'Equals' or 'exactly' points to an equation (=); 'at least, at most, more than, fewer than' points to an inequality." — always check your work against this rule before you submit.*
8. **Try It 1:** A. $x \le 20$ — *'At most 20' means 20 or fewer, so $x \le 20$.*
9. **Try It 2:** A. $w = 12$ — *Divide both sides by 5: $w = 60 \div 5 = 12$ pounds. Check: $5 \times 12 = 60$ ✓*
10. **Exit Ticket:** A. $d - 7 < 5$ — *Starting with d donuts and giving away 7 leaves fewer than 5: $d - 7 < 5$. Solve: $d < 12$. ✓*